CryptoSSU

Advanced Molecular
Detection Southeast Region

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UPDATES

Moving to once-a-month meeting occurrence!

OVERVIEW

Purpose:

- This Nextflow pipeline processes Illumina sequencing reads to detect and identify *Cryptosporidium* species and subtypes, generating outputs such as cleaned reads, assemblies, and summary results.

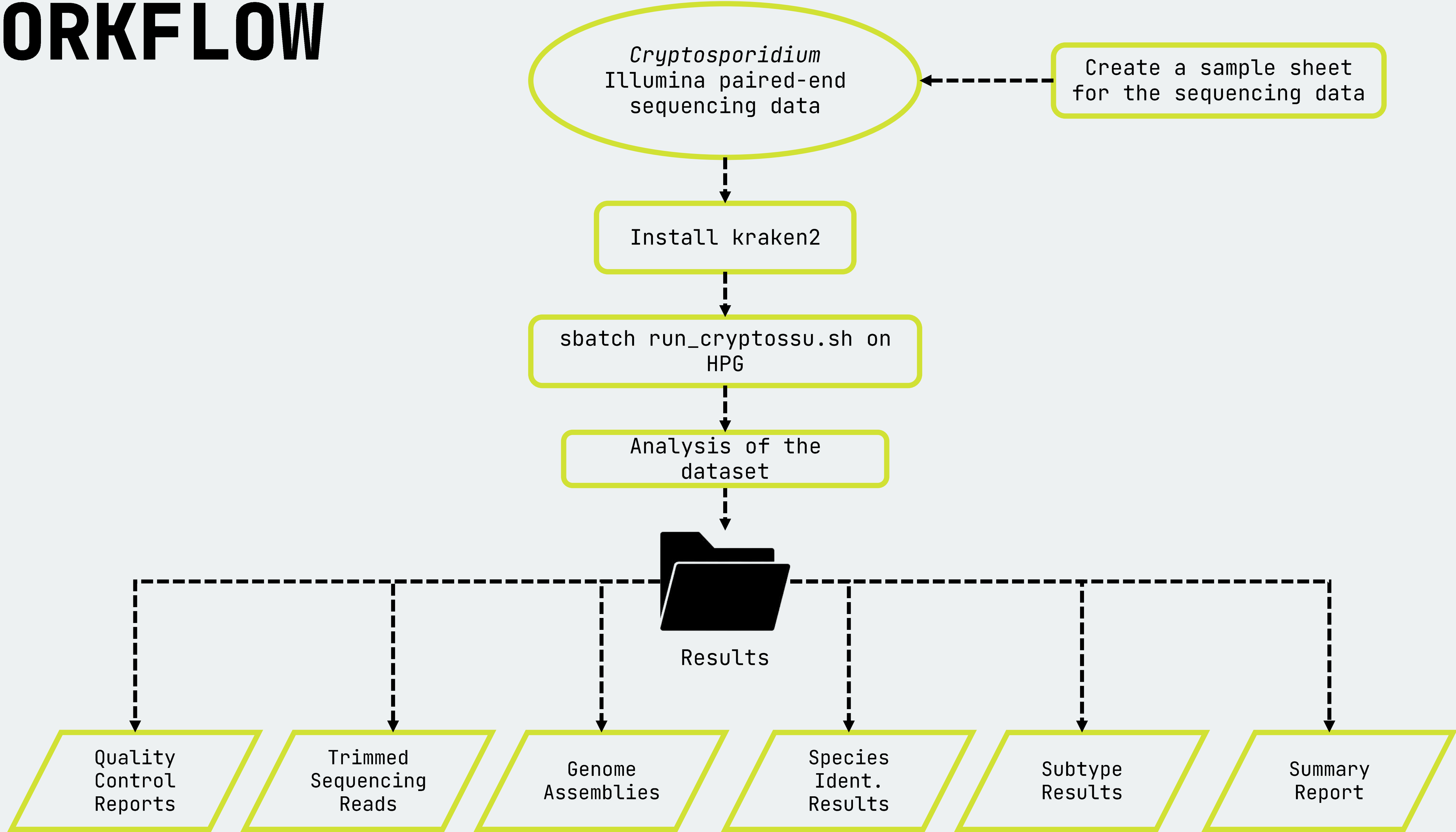
Usage:

- It helps public health labs and researchers identify Cryptosporidium infections and analyze them for surveillance and outbreak tracking.

Dependencies:

- Nextflow
- Java
- Apptainer/Singularity
- Kraken2

WORKFLOW


Cryptosporidium Illumina paired-end sequencing data
Create a sample sheet for the sequencing data
Install kraken2
sbatch run_cryptossu.sh on HPG
Analysis of the dataset
Results
Quality Control Reports
Trimmed Sequencing Reads
Genome Assemblies
Species Ident. Results
Subtype Results
Summary Report

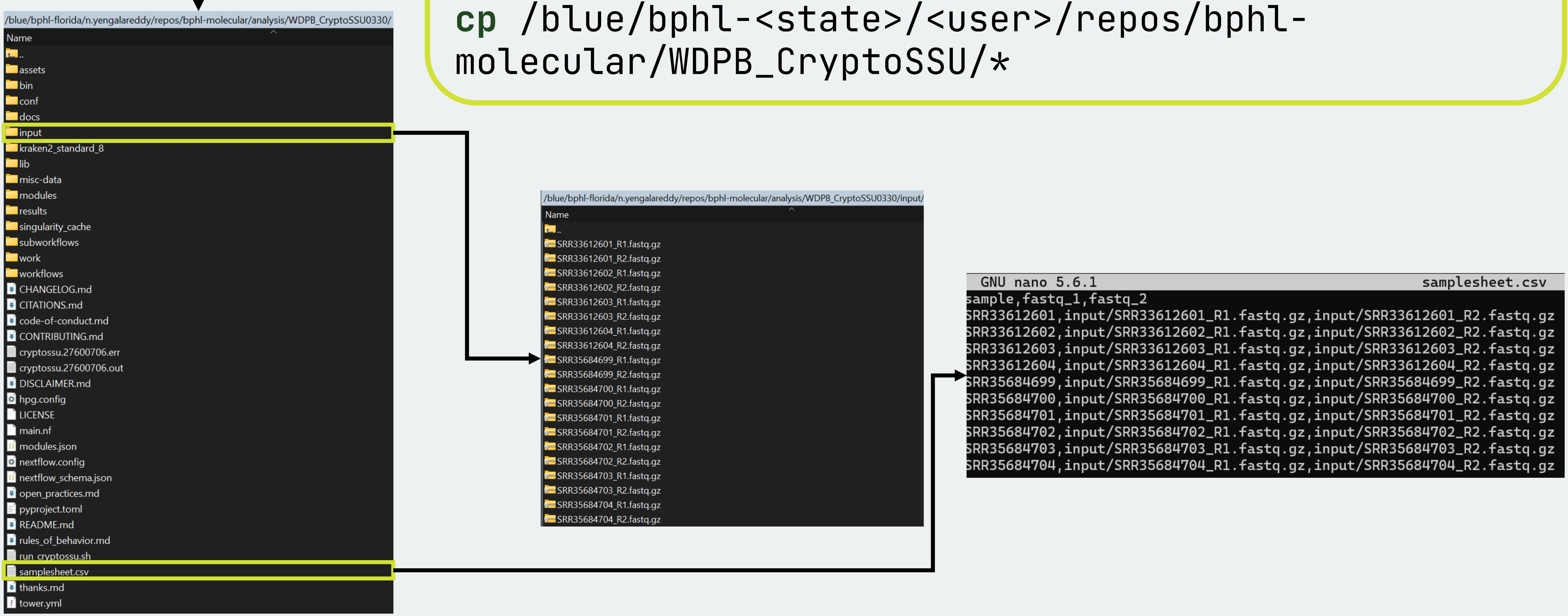
Application

OBJECTIVE

Use a *Cryptosporidium* dataset and analyze the dataset using CryptoSSU pipeline

Application Cont.

```
cd /blue/bphl-<state>/<user>/repos/bphl-molecular/
git clone https://github.com/CDCgov/WDPB_CryptoSSU
mkdir analysis/
cd analysis/
cp /blue/bphl-<state>/<user>/repos/bphl-
molecular/WDPB_CryptoSSU/*
```



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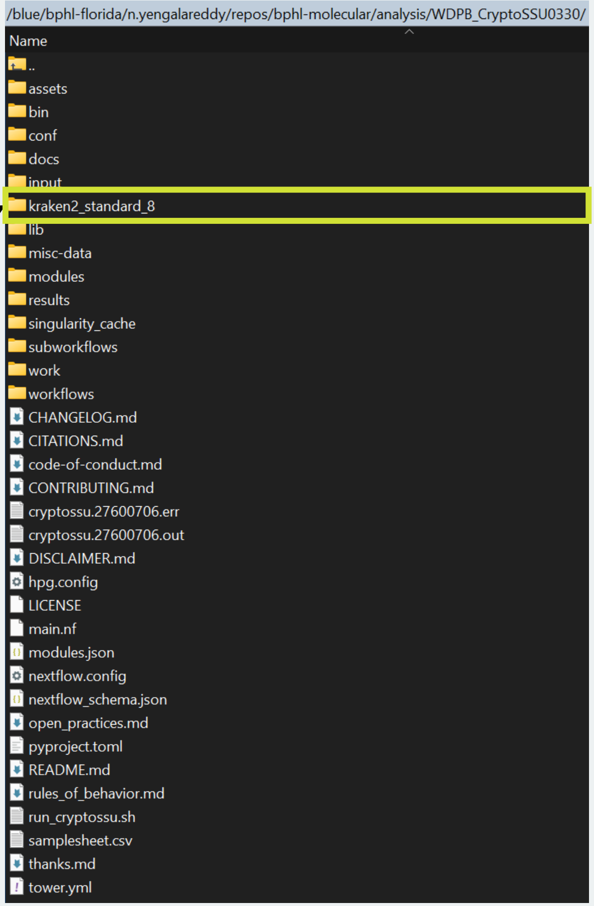
Southeast Region Bioinformatics

Application Cont.

```
cd /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/WDPB_CryptoSSU

mkdir kraken2_standard_8/

wget https://genome-idx.s3.amazonaws.com/kraken/k2_standard_08gb_20250402.tar.gz

tar -xvzf k2_standard_08gb_20250402.tar.gz
```

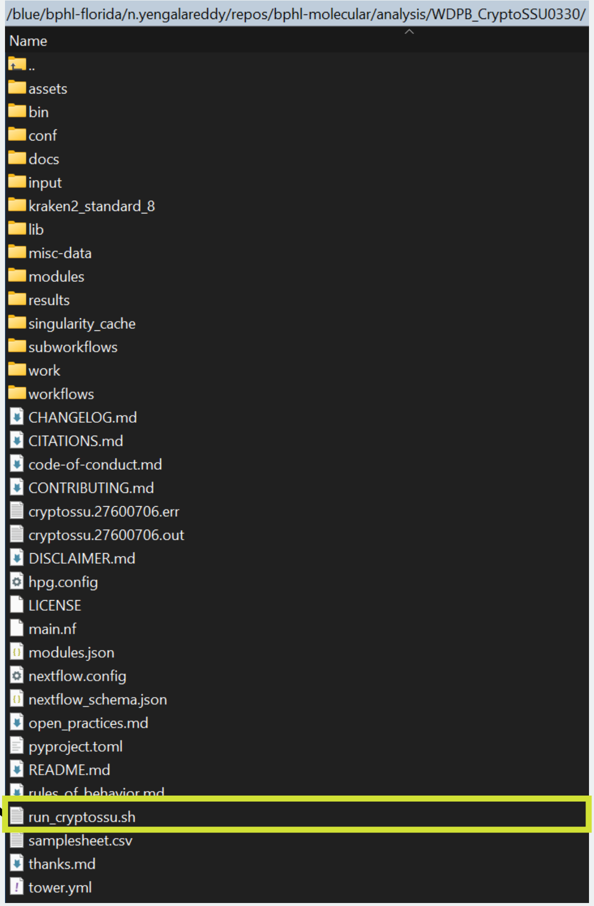
Delete k2_standard_08gb_20250402.tar.gz after

Check for hash.k2d, opts.k2d, and taxo.k2d files



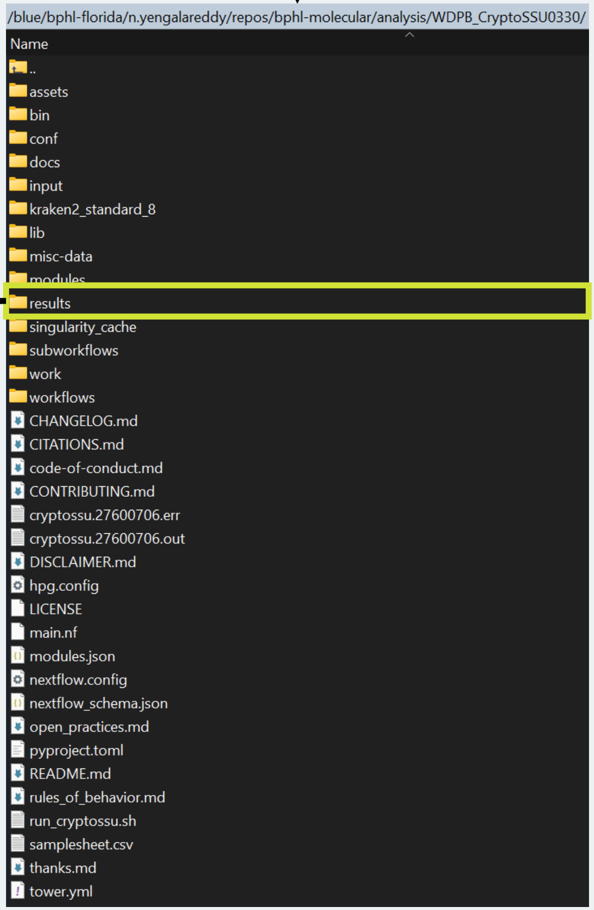
Application Cont.

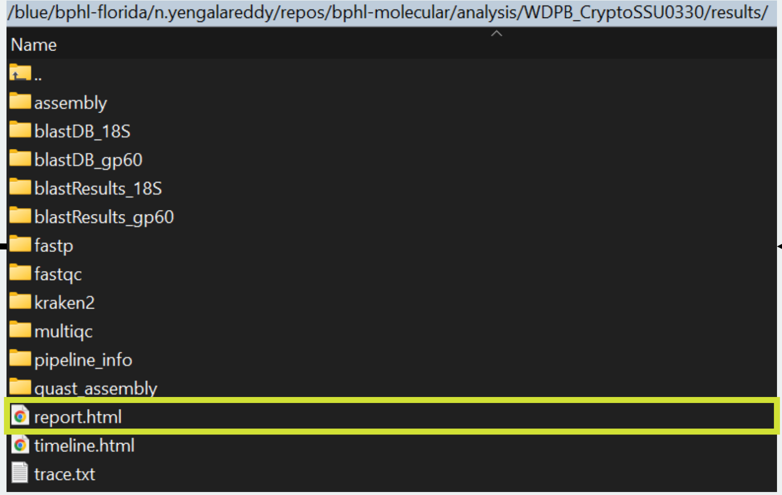
```
GNU nano 5.6.1                                                                  run_cryptossu.sh
#!/bin/bash
#SBATCH --account=bphl-umbrella
#SBATCH --qos=bphl-umbrella
#SBATCH --job-name=cryptossu
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=16
#SBATCH --mem=160G
#SBATCH --time=24:00:00
#SBATCH --output=cryptossu.%j.out
#SBATCH --error=cryptossu.%j.err
#SBATCH --mail-type=END,FAIL
#SBATCH --mail-user=nikhil.yengala@flhealth.gov

module purge
module load nextflow
module load singularity

BASE=/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/WDPB_CryptoSSU0330

WORKDIR=${BASE}/work
OUTDIR=${BASE}/results
SINGCACHE=${BASE}/singularity_cache
SAMPLESHEET=${BASE}/samplesheet.csv
CONFIG=${BASE}/hpg.config

KRAKEN_DB=${BASE}/kraken2_standard_8

mkdir -p "$WORKDIR"
mkdir -p "$OUTDIR"
mkdir -p "$SINGCACHE"

export NXF_SINGULARITY_CACHEDIR="$SINGCACHE"
export NXF_OPTS='-Xms4g -Xmx32g'

nextflow run CDCgov/WDPB_CryptoSSU \
  -r main \
  -profile singularity \
  -c "$CONFIG" \
  -work-dir "$WORKDIR" \
  --input "$SAMPLESHEET" \
  --outdir "$OUTDIR" \
  --platform illumina \
  --allele_seq "$BASE/assets/allalleles_0.fasta" \
  --kraken2_db "$KRAKEN_DB" \
  -resume \
  -with-report "$OUTDIR/report.html" \
  -with-timeline "$OUTDIR/timeline.html" \
  -with-trace "$OUTDIR/trace.txt"
```

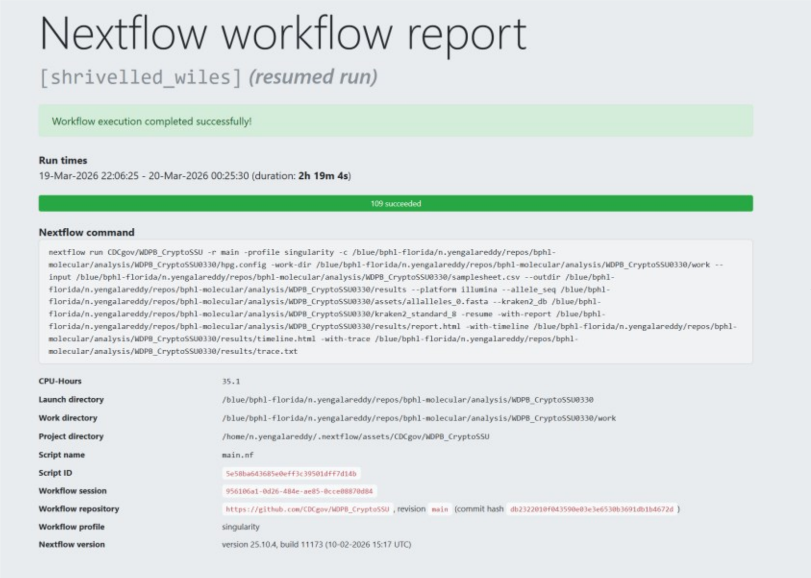


Application Cont.

`sbatch run_cryptossu.sh`







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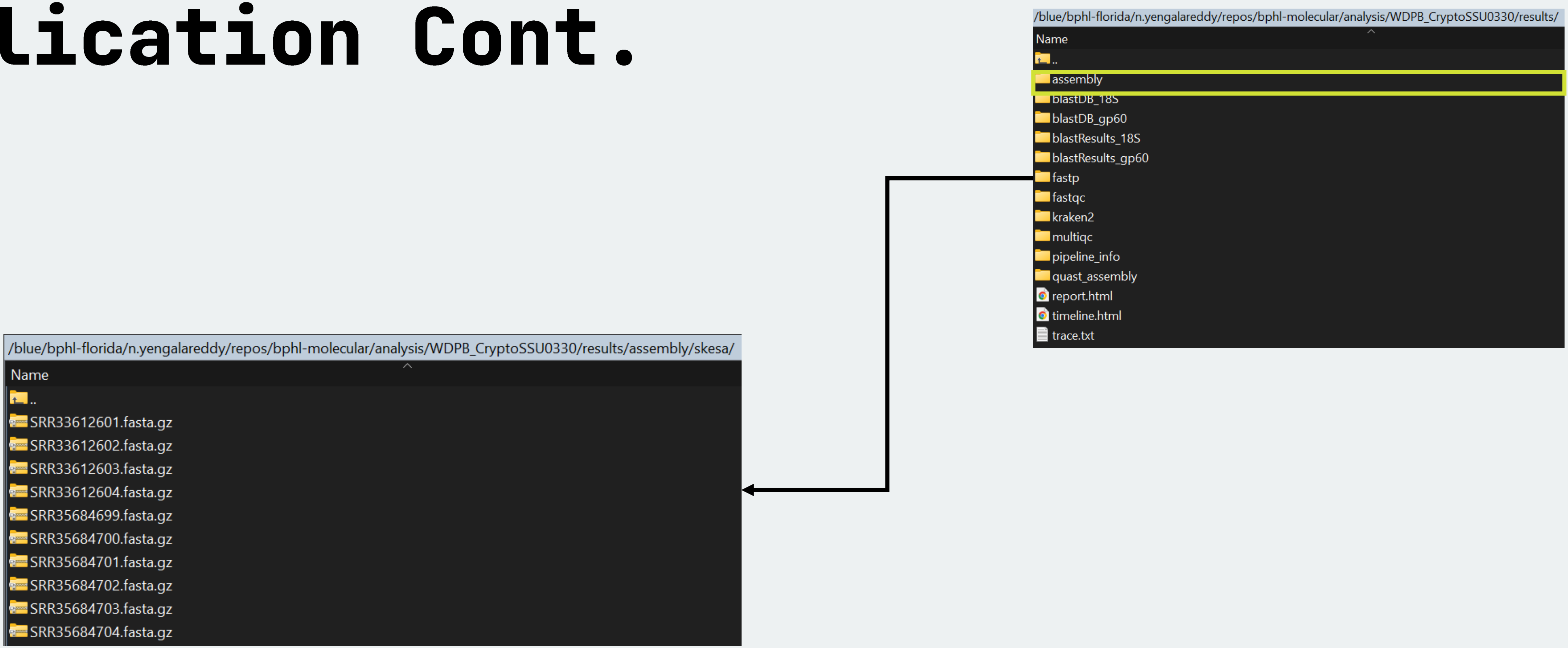
Application Cont.



Application Cont.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/WDPB_CryptoSSU0330/results/

Name

- ..
- assembly
- blastDB_18S
- blastDB_gp60
- blastResults_18S
- blastResults_gp60
- fastp
- fastqc
- kraken2
- multiqc
- pipeline_info
- quast_assembly
- report.html
- timeline.html
- trace.txt

/.../repos/bphl-molecular/analysis/WDPB_CryptoSSU0330/results/blastDB_18S/DB_18S_Crypto_20241120.fasta_DB/

Name

- ..
- DB_18S_Crypto_20241120.fasta.ndb
- DB_18S_Crypto_20241120.fasta.nhr
- DB_18S_Crypto_20241120.fasta.nin
- DB_18S_Crypto_20241120.fasta.njs
- DB_18S_Crypto_20241120.fasta.not
- DB_18S_Crypto_20241120.fasta.nsq
- DB_18S_Crypto_20241120.fasta.ntf
- DB_18S_Crypto_20241120.fasta.nto

Advanced Molecular Detection
Southeast Region Bioinformatics

Application Cont.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/WDPB_CryptoSSU0330/results/

Name
- ..
- assembly
- blastDB_18S
- blastDB_gp60
- blastResults_18S
- blastResults_gp60
- fastp
- fastqc
- kraken2
- multiqc
- pipeline_info
- quast_assembly
- report.html
- timeline.html
- trace.txt

/.../repos/bphl-molecular/analysis/WDPB_CryptoSSU0330/results/blastDB_gp60/DB_GP60_Crypto_20241120.fasta_DB/

Name
- ..
- DB_GP60_Crypto_20241120.fasta.ndb
- DB_GP60_Crypto_20241120.fasta.nhr
- DB_GP60_Crypto_20241120.fasta.nin
- DB_GP60_Crypto_20241120.fasta.njs
- DB_GP60_Crypto_20241120.fasta.not
- DB_GP60_Crypto_20241120.fasta.nsq
- DB_GP60_Crypto_20241120.fasta.ntf
- DB_GP60_Crypto_20241120.fasta.nto

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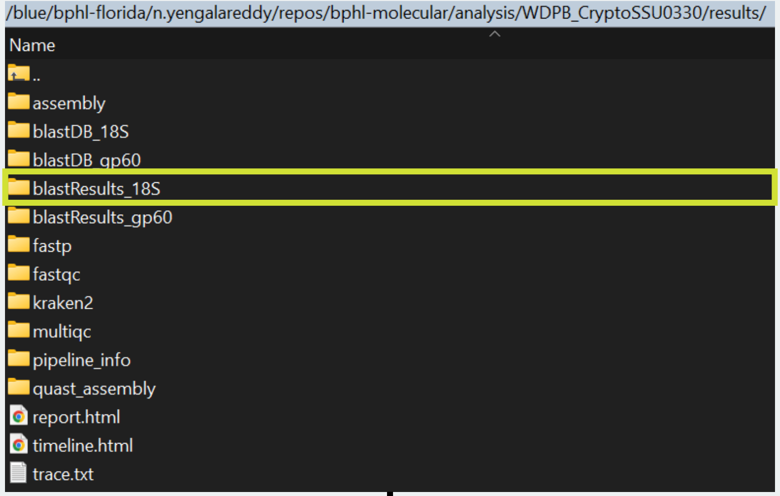
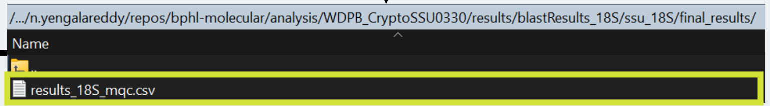
Southeast Region Bioinformatics

Application Cont.

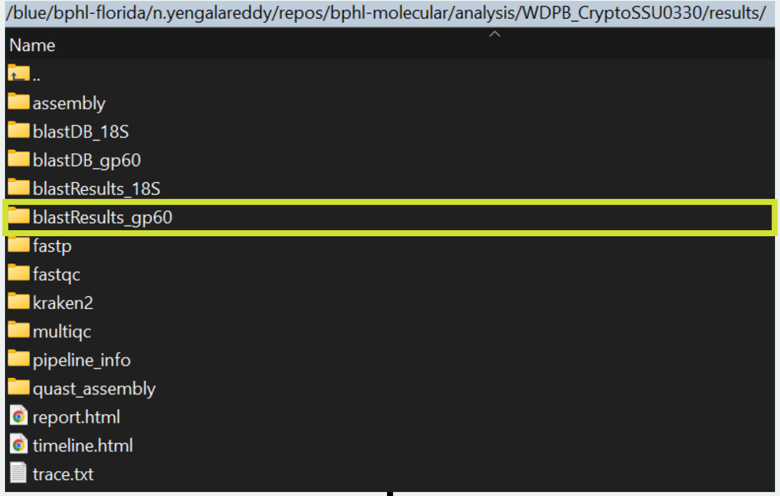


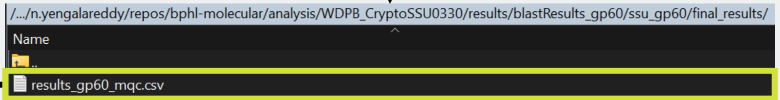
Sample Name	query_genome	db_bestmatch	species	pident	alignment_length	coverage	query_start	query_end	subject_start	subject_end	bitscore	NCE
SRR35684699	Contig_184_250.429	C_parvum_AAEE01000005.299599.301356_GENOTYPE_NAN_18s	C.parvum	100	1758	2	11815	13572	1758	1	3247	NA;NA
SRR35684701	Contig_865_168.664	C_parvum_AAEE01000005.299599.301356_GENOTYPE_NAN_18s	C.parvum	100	1758	8	11816	13573	1758	1	3247	NA;NA
SRR35684704	Contig_1300_236.435	C_parvum_AAEE01000005.299599.301356_GENOTYPE_NAN_18s	C.parvum	100	1758	6	17822	19579	1758	1	3247	NA;NA
SRR33612601	Contig_162_213.679	C_parvum_AAEE01000005.299599.301356_GENOTYPE_NAN_18s	C.parvum	100	1758	2	17778	19535	1758	1	3247	NA;NA
SRR35684703	Contig_157_172.189	C_parvum_AAEE01000005.299599.301356_GENOTYPE_NAN_18s	C.parvum	100	1758	4	17778	19535	1758	1	3247	NA;NA
SRR35684700	Contig_836_222.191	C_parvum_AAEE01000005.299599.301356_GENOTYPE_NAN_18s	C.parvum	100	1758	2	11835	13592	1758	1	3247	NA;NA
SRR33612602	Contig_771_206.091	C_parvum_AAEE01000005.299599.301356_GENOTYPE_NAN_18s	C.parvum	100	1758	2	45513	47270	1758	1	3247	NA;NA
SRR33612604	Contig_875_243.835	C_parvum_AAEE01000005.299599.301356_GENOTYPE_NAN_18s	C.parvum	100	1758	3	5484	7241	1758	1	3247	NA;NA
SRR35684702	Contig_181_312.671	C_parvum_AAEE01000005.299599.301356_GENOTYPE_NAN_18s	C.parvum	100	1758	2	17779	19536	1758	1	3247	NA;NA
SRR33612603	Contig_161_216.476	C_parvum_AAEE01000005.299599.301356_GENOTYPE_NAN_18s	C.parvum	100	1758	2	17779	19536	1758	1	3247	NA;NA



Advanced Molecular Detection
Southeast Region Bioinformatics

Application Cont.





Sample Name	gp60_Subtype	Fasta_Header	Database_Best_Match	Length(bp)	Blast_Identity	Coverage	NCE
SRR35684701	IIdA20G1	Contig_23_494.172	C_parvum_IIdA23G1_37-109_JF727762_Spain_cow	1291	97.964	68	
SRR35684699	IIdA20G1	Contig_954_261.094	C_parvum_IIdA23G1_37-109_JF727762_Spain_cow	109807	97.964	1	
SRR35684704	IIaA17G2R1	Contig_230_722.279	C_parvum_IIaA17G2R1_51-113_49498	854	99.649	100	
SRR33612601	IIdA20G1	Contig_1037_718.652	C_parvum_IIdA23G1_37-109_JF727762_Spain_cow	993	97.943	87	
SRR35684703	IIdA20G1	Contig_697_187.144	C_parvum_IIdA23G1_37-109_JF727762_Spain_cow	456850	97.964	0	
SRR35684700	IIdA20G1	Contig_514_301.779	C_parvum_IIdA23G1_37-109_JF727762_Spain_cow	150178	97.964	1	
SRR33612602	IIdA20G1	Contig_980_230.349	C_parvum_IIdA23G1_37-109_JF727762_Spain_cow	462066	97.964	0	
SRR33612604	IIaA17G2R1	Contig_1358_748.269	C_parvum_IIaA16G3R1_51-113_49518	1005	99.535	86	
SRR35684702	IIdA20G1	Contig_567_405.064	C_parvum_IIdA23G1_37-109_JF727762_Spain_cow	165742	97.964	1	
SRR33612603	IIdA20G1	Contig_690_238.781	C_parvum_IIdA23G1_37-109_JF727762_Spain_cow	516206	97.964	0	

Advanced Molecular Detection
Southeast Region Bioinformatics

Application Cont.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/WDPB_CryptoSSU0330/results/

Name

- ..
- assembly
- blastDB_18S
- blastDB_gp60
- blastResults_18S
- blastResults_gp60
- fastp
- fastqc
- kraken2
- multiqc
- pipeline_info
- quast_assembly
- report.html
- timeline.html
- trace.txt

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/WDPB_CryptoSSU0330/results/fastp/

Name

- ..
- SRR33612601.fastp.html
- SRR33612601.fastp.json
- SRR33612601.fastp.log
- SRR33612601_1.fastp.fastq.gz
- SRR33612601_2.fastp.fastq.gz
- SRR33612602.fastp.html
- SRR33612602.fastp.json
- SRR33612602.fastp.log
- SRR33612602_1.fastp.fastq.gz
- SRR33612602_2.fastp.fastq.gz

fastp report

Summary

General

fastp version:	0.23.2 (https://github.com/OpenGene/fastp)
sequencing:	paired end (151 cycles + 151 cycles)
mean length before filtering:	144bp, 144bp
mean length after filtering:	144bp, 144bp
duplication rate:	9.271556%
Insert size peak:	201

Before filtering

total reads:	17.054462 M
total bases:	2.470961 G
Q20 bases:	2.370629 G (95.939538%)
Q30 bases:	2.240708 G (90.681623%)
GC content:	31.673569%

After filtering

total reads:	16.973490 M
total bases:	2.459065 G
Q20 bases:	2.362298 G (96.064904%)
Q30 bases:	2.233730 G (90.836544%)
GC content:	31.665529%

Filtering result

reads passed filters:	16.973490 M (99.525215%)
reads with low quality:	71.332000 K (0.418260%)
reads with too many N:	9.640000 K (0.056525%)
reads too short:	0 (0.000000%)

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Southeast Region Bioinformatics

Application Cont.

FastQC Report

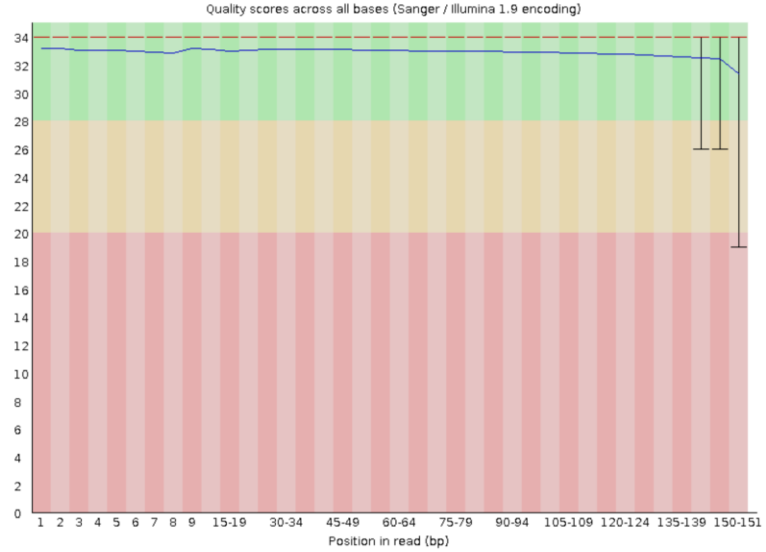
Summary

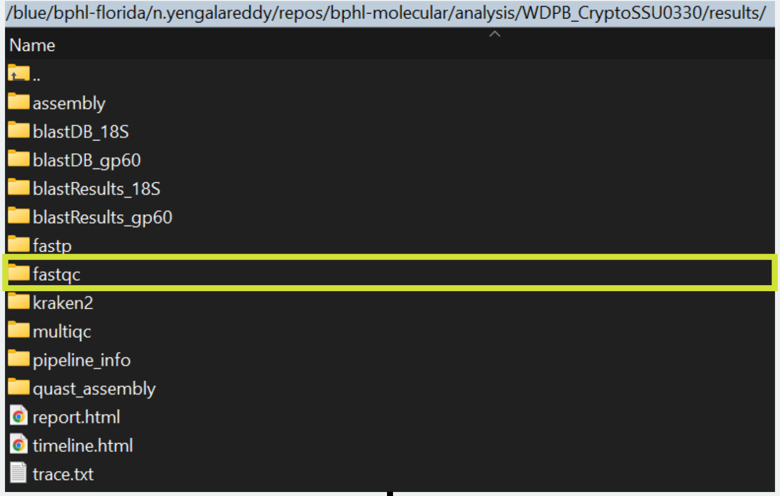
- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

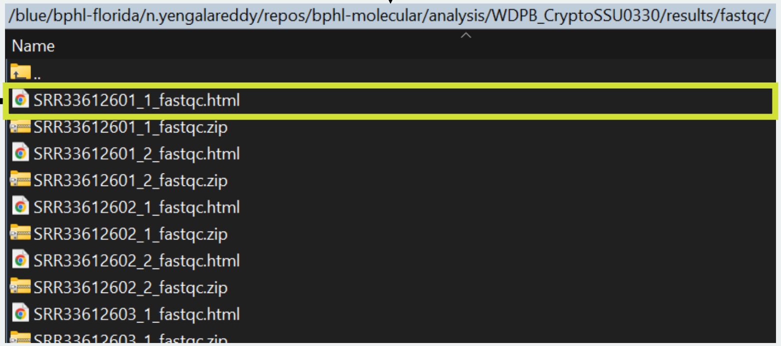
Basic Statistics

Measure	Value
Filename	SRR33612601_1.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	8486745
Total Bases	1.2 Gbp
Sequences flagged as poor quality	0
Sequence length	31-151
%GC	31

Per base sequence quality







Advanced Molecular Detection
Southeast Region Bioinformatics

Application Cont.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/WDPB_CryptoSSU0330/results/

Name
- ..
- assembly
- blastDB_18S
- blastDB_gp60
- blastResults_18S
- blastResults_gp60
- fastp
- fastqc
- kraken2
- multiqc
- pipeline_info
- quast_assembly
- report.html
- timeline.html
- trace.txt

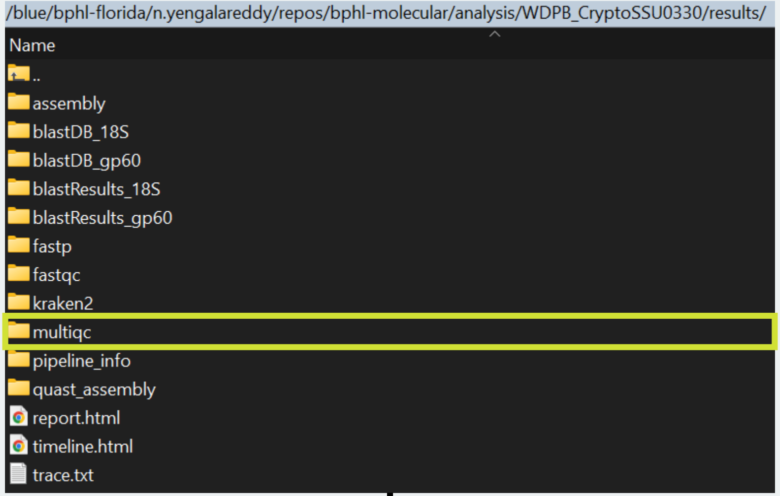
/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/WDPB_CryptoSSU0330/results/kraken2/

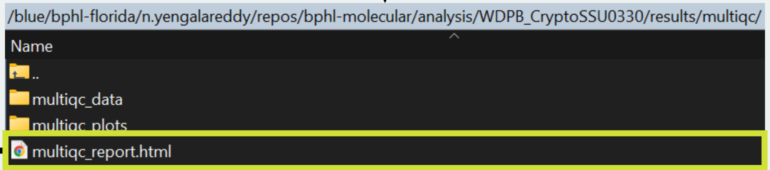
Name
- ..
- SRR33612601.kraken2.report.txt
- SRR33612602.kraken2.report.txt
- SRR33612603.kraken2.report.txt
- SRR33612604.kraken2.report.txt
- SRR35684699.kraken2.report.txt
- SRR35684700.kraken2.report.txt
- SRR35684701.kraken2.report.txt
- SRR35684702.kraken2.report.txt
- SRR35684703.kraken2.report.txt
- SRR35684704.kraken2.report.txt

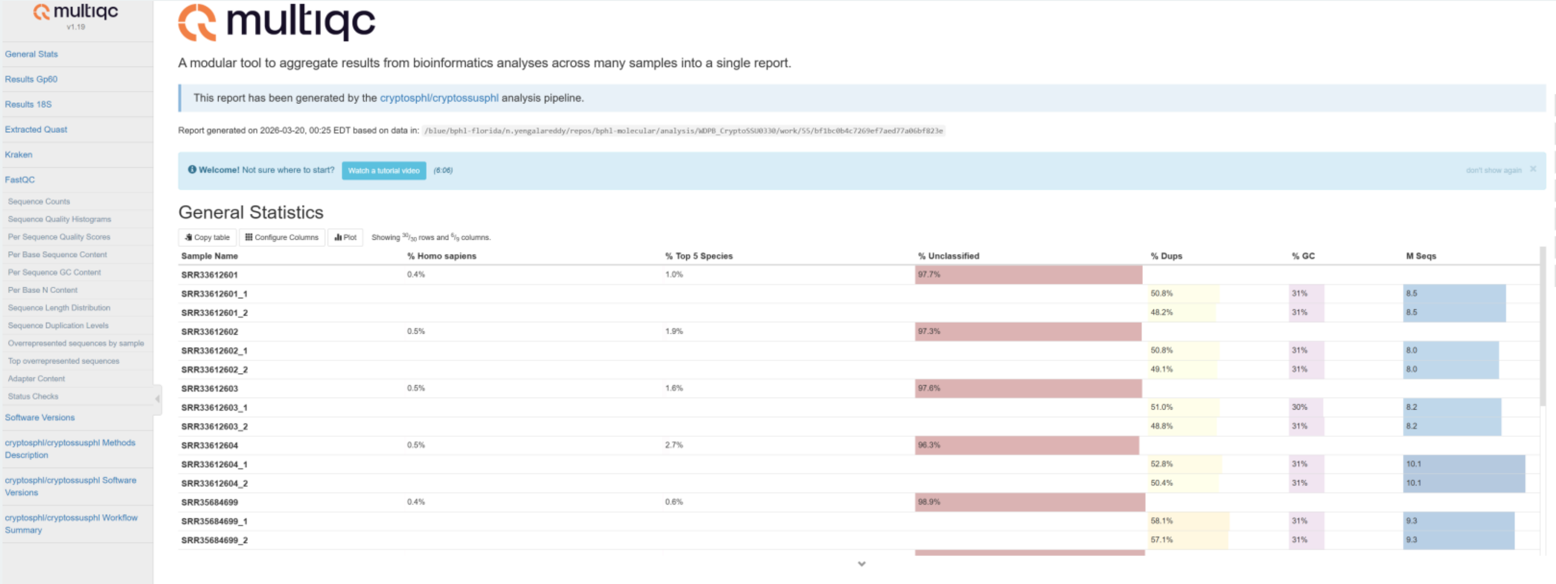
```
97.69  8290340 8290340 U   0        unclassified
 2.31  196405  135     R   1        root
 2.31  195725  2765    R1  131567     cellular organisms
 1.83  155017  5056    R2  2            Bacteria
 1.29  109758  436     K   1783272        Bacillati
 0.77  65612   104     P   201174           Actinomycetota
 0.52  44410   224     C   1760               Actinomycetes
 0.48  41076   0       O   85004                Bifidobacteriales
 0.48  41076   58      F   31953                  Bifidobacteriaceae
 0.48  41015   1058    G   1678                     Bifidobacterium
 0.47  39632   34845   S   1694                       Bifidobacterium pseudolongum
 0.03  2187    1189    S1  1690                         Bifidobacterium pseudolongum subsp. globosum
 0.01  998     998     S2  1410605                        Bifidobacterium pseudolongum subsp. globosum DSM 20092
 0.03  2186    2186    S1  1447715                      Bifidobacterium pseudolongum PV8-2
 0.00  414     414     S1  31954                        Bifidobacterium pseudolongum subsp. pseudolongum
 0.00  213     174     S   28025                      Bifidobacterium animalis
 0.00  39      39      S1  302911                       Bifidobacterium animalis subsp. lactis
 0.00  42      42      S   35760                      Bifidobacterium choerinum
 0.00  37      32      S   216816                     Bifidobacterium longum
 0.00  5       2       S1  1682                         Bifidobacterium longum subsp. infantis
 0.00  3       3       S2  565040                         Bifidobacterium longum subsp. infantis 157F
 0.00  11      0       S   78448                      Bifidobacterium pullorum
 0.00  11      11      S1  78344                        Bifidobacterium pullorum subsp. gallinarum
 0.00  3       3       S   1683                       Bifidobacterium angulatum
 0.00  3       3       S   1684                       Bifidobacterium asteroides
 0.00  2       2       S   1680                       Bifidobacterium adolescentis
 0.00  2       2       S   1765219                    Bifidobacterium eulemuris
 0.00  2       2       S   2020965                    Bifidobacterium imperatoris
 0.00  2       0       S   158787                     Bifidobacterium scardovii
 0.00  2       2       S1  1150461                      Bifidobacterium scardovii JCM 12489 = DSM 13734
 0.00  2       1       S   1686                       Bifidobacterium catenulatum
 0.00  1       0       S1  630129                       Bifidobacterium catenulatum subsp. kashiwanohense
 0.00  1       1       S2  1150460                        Bifidobacterium catenulatum subsp. kashiwanohense JCM 15439 = DSM 21854
 0.00  1       1       S   33905                      Bifidobacterium thermophilum
 0.00  1       1       S   28026                      Bifidobacterium pseudocatenulatum
 0.00  1       1       S   77635                      Bifidobacterium subtile
 0.00  1       1       G1  2608897                    unclassified Bifidobacterium
 0.00  1       1       S   1685                       Bifidobacterium breve
 0.00  1       1       S   762210                     Bifidobacterium saguini
 0.00  2       0       G   196082                   Parascardovia
 0.00  2       0       S   78258                      Parascardovia denticolens
 0.00  2       2       S1  864564                       Parascardovia denticolens DSM 10105 = JCM 12538
 0.00  1       0       G   196081                   Scardovia
 0.00  1       0       S   78259                      Scardovia inopinata
 0.00  1       1       S1  1150468                      Scardovia inopinata JCM 12537
 0.02  1537    0       O   85011                Kitasatosporales
 0.02  1537    2       F   2062                   Streptomycetaceae
 0.02  1478    150     G   1883                     Streptomyces
 0.01  1104    248     G1  2593676                    unclassified Streptomyces
 0.00  203     203     S   2903788                      Streptomyces sp. NBC_01235
 0.00  146     146     S   3050126                      Streptomyces sp. KL2
 0.00  53      53      S   2903700                      Streptomyces sp. NBC_00984
 0.00  46      46      S   2903749                      Streptomyces sp. NBC_01102
 0.00  42      42      S   2903885                      Streptomyces sp. NBC_01497
 0.00  41      41      S   2975717                      Streptomyces sp. NBC_00341
 0.00  37      37      S   2975742                      Streptomyces sp. NBC_00441
 0.00  36      36      S   3231512                      Streptomyces sp. CMC78
 0.00  36      36      S   3240934                      Streptomyces sp. QH1-20
 0.00  31      31      S   2692234                      Streptomyces sp. GS7
 0.00  29      29      S   2903665                      Streptomyces sp. NBC_00576
 0.00  29      29      S   1851167                      Streptomyces sp. 11-1-2
 0.00  26      26      S   2975917                      Streptomyces sp. NBC_01716
 0.00  24      24      S   2903755                      Streptomyces sp. NBC_01166
 0.00  9       9       S   2136173                      Streptomyces sp. So13.3
 0.00  5       5       S   2975946                      Streptomyces sp. NBC_01803
 0.00  5       5       S   3372854                      Streptomyces sp. BBFR2
 0.00  4       4       S   2903862                      Streptomyces sp. NBC_01429
 0.00  3       3       S   3388313                      Streptomyces sp. CY1
```

Advanced Molecular Detection
Southeast Region Bioinformatics

Application Cont.







Application Cont.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/WDPB_CryptoSSU0330/results/

Name
..
assembly
blastDB_18S
blastDB_gp60
blastResults_18S
blastResults_gp60
fastp
fastqc
kraken2
multiqc
pipeline_info
quast_assembly
report.html
timeline.html
trace.txt

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/WDPB_CryptoSSU0330/results/quast_assembly/quast/

Name
..
aligned_stats
basic_stats
contigs_reports
genome_stats
icarus_viewers
icarus.html
quast.log
report.html
report.pdf
report.tex
report.tsv
report.txt
transposed_report.tex
transposed_report.tsv
transposed_report.txt

Report

	SRR35684701	SRR35684699	SRR35684704	SRR33612601	SRR35684703	SRR35684700	SRR33612602	SRR33612604	SRR35684702	SRR33612603
# contigs (>= 0 bp)	1223	1146	1319	1243	1211	1175	1618	1619	1284	1176
# contigs (>= 1000 bp)	948	894	989	982	926	906	1125	1215	969	900
# contigs (>= 5000 bp)	443	389	457	470	425	399	417	517	408	418
# contigs (>= 10000 bp)	232	207	223	258	211	207	174	231	184	206
# contigs (>= 25000 bp)	60	62	59	66	66	65	55	44	61	50
# contigs (>= 50000 bp)	17	29	25	15	22	22	21	11	24	23
Total length (>= 0 bp)	9075501	9082701	9066772	9103037	9077546	9081712	9088252	9117716	9068209	9122698
Total length (>= 1000 bp)	8906997	8936514	8867437	8947563	8917828	8921135	8809350	8884137	8886340	8961295
Total length (>= 5000 bp)	7607955	7670584	7524289	7652797	7641875	7672349	7069242	7101838	7437365	7690328
Total length (>= 10000 bp)	6096328	6364908	5851997	6140379	6143539	6292970	5355070	5053464	5834565	6172153
Total length (>= 25000 bp)	3408015	4191875	3398663	3269753	3865945	4099338	3521466	2329028	4078948	3785106
Total length (>= 50000 bp)	1863304	3085245	2244169	1489991	2360048	2657368	2340722	1180005	2772731	2861204
# contigs	1125	1037	1195	1141	1088	1072	1404	1447	1152	1061
Largest contig	255699	254295	229951	283085	456850	318860	462066	215373	301361	516206
Total length	9039133	9043111	9021100	9066298	9034153	9043365	9013313	9057357	9021382	9082998
Reference length	9102324	9102324	9102324	9102324	9102324	9102324	9102324	9102324	9102324	9102324
GC (%)	30.18	30.19	30.20	30.22	30.20	30.20	30.27	30.26	30.19	30.24
Reference GC (%)	30.22	30.22	30.22	30.22	30.22	30.22	30.22	30.22	30.22	30.22
N50	17496	21716	16620	15806	19725	21714	15193	11180	18630	19021
NG50	17324	21367	16204	15803	18878	21582	14756	11154	17812	18847
N90	3484	3439	3219	3408	3546	3402	2585	2695	3256	3559
NG90	3354	3282	3047	3316	3423	3274	2408	2599	3088	3498
auN	44214.1	56132.9	37662.2	35150.5	58295.9	58761.7	57716.5	29188.7	55541.9	73133.5
auNG	43907.2	55767.7	37326.1	35011.3	57859.3	58381.1	57152.1	29044.5	55048.0	72978.3
L50	115	77	116	131	96	84	106	182	82	86
LG50	116	78	119	132	98	85	109	184	84	87
L90	571	501	603	592	541	513	704	801	574	532
LG90	587	517	626	601	558	529	736	816	597	537
# misassemblies	5	4	7	6	6	5	4	5	6	7
# misassembled contigs	5	4	7	6	6	5	4	5	6	7
Misassembled contigs length	170872	88120	121859	78768	169997	182719	47263	56778	146828	248159
# local misassemblies	3	4	4	6	3	5	3	3	5	6
# scaffold gap ext. mis.	0	0	0	0	0	0	0	0	0	0
# scaffold gap loc. mis.	0	0	0	0	0	0	0	0	0	0
# unaligned mis. contigs	0	0	0	1	0	0	1	2	0	2
# unaligned contigs	12 + 23 part	12 + 22 part	9 + 24 part	31 + 22 part	16 + 23 part	10 + 22 part	48 + 27 part	50 + 24 part	15 + 21 part	33 + 24 part
Unaligned length	87195	80572	81136	105802	82747	78683	123283	132142	83573	114607
Genome fraction (%)	98.237	98.326	98.121	98.354	98.215	98.365	97.569	97.896	98.152	98.424
Duplication ratio	1.003	1.003	1.003	1.003	1.003	1.003	1.003	1.003	1.002	1.003
# N's per 100 kbp	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
# mismatches per 100 kbp	27.47	33.69	36.27	15.72	38.32	34.69	33.63	18.96	33.96	40.56
# indels per 100 kbp	12.99	14.68	14.95	8.48	15.64	13.59	15.47	10.30	14.96	17.51
# genomic features	17750 + 2620 part	18166 + 2216 part	17725 + 2647 part	17680 + 2716 part	17916 + 2454 part	17947 + 2423 part	17248 + 3074 part	17206 + 3171 part	17847 + 2539 part	18053 + 2345 part
Largest alignment	255699	254295	228482	283085	456850	318860	462066	215373	301361	516206
Total aligned length	8948008	8958780	8936269	8959128	8947543	8961235	8888096	8922972	8934908	8967402
NA50	17188	21193	16363	15710	18775	21209	14750	11154	17478	18506
NGA50	17091	21080	15780	15685	18750	20898	14373	11095	16865	18325
NA90	3254	3210	2994	3211	3316	3206	2389	2499	3080	3304
NGA90	3043	3025	2808	3034	3091	3018	2164	2416	2850	3206
auNA	43522.9	55755.4	37316.4	34758.3	57890.4	58247.1	57519.3	29068.8	55176.9	71644.4
auNGA	43220.8	55392.7	36983.4	34620.7	57456.8	57869.8	56956.8	28925.2	54686.2	71492.3
LA50	117	78	118	133	98	85	107	183	84	88
LGA50	119	79	121	134	100	87	110	185	86	89
LA90	592	519	625	608	560	534	729	823	594	555
LGA90	610	536	650	619	579	551	764	839	618	560

All statistics are based on contigs of size >= 500 bp, unless otherwise noted
(e.g., "# contigs (>= 0 bp)" and "Total length (>= 0 bp)" include all contigs).

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Conclusion

- Fundamentals of CryptoSSU
- Installation and setup of CryptoSSU in HPG
- Successfully executed job query for CryptoSSU
- Generated output files

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Questions?

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